Logically Let us C

Computer Science & IT

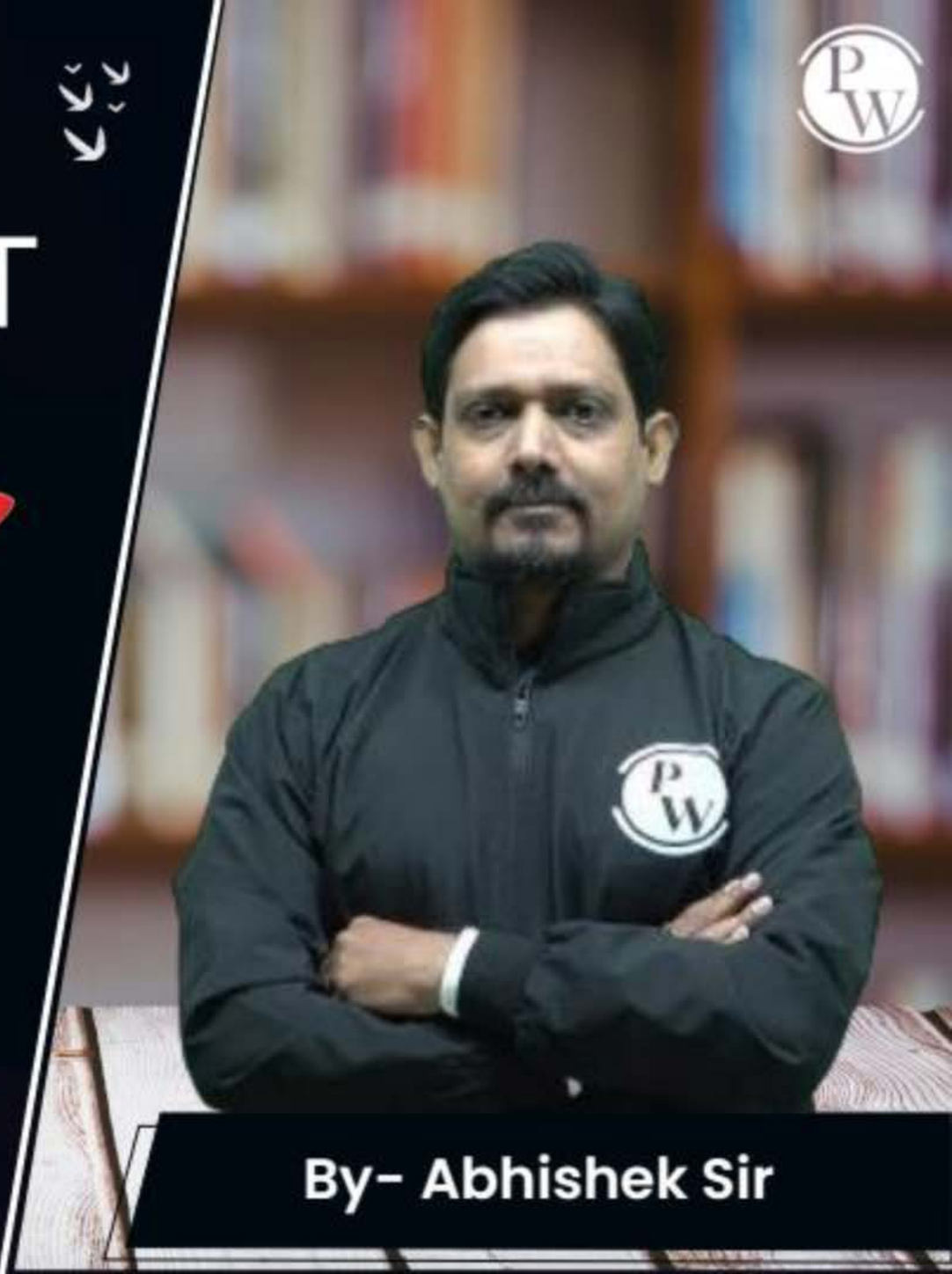
Engineer : problem



# C Programming

## Control Flow Statements

Lecture No. 04



By- Abhishek Sir

# Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Switch

loop

start

for loop

Slide

# Topics to be Covered



- Topic — For Loops
- Topic
- Topic
- Topic
- Topic

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# For loop

```
for(expr1; expr2; expr3) {
```

No semicolon

if Single stmt the if block is Not created it's ok

}

multiple statement then create a block



# For loop

exp$_1$ ← one time execute. ( Assignment)

exp$_2$ ← Condition if condition true (as it was in if the block will execute.

zero false

Non zero true.

exp$_3$ ← Increment / Decrement

Slide



# For loop

```
int i;
for (i=1; i<=3; i++) {
    printf("I am a good student");
}
```

one time

Iteration 1 : 1<=3, I am.... ✓

i=2 Iteration 2 : 2<=3, I am.... ✓

i=3, Iteration 3 : 3<=3, I am.... ✓

i=4 Iteration 4 : 4<=3 ← false then it will exit.

Slide



# For loop

expr3 always executed after executing all statement within the block

Execute - print / evaluate.

Slide



# For loop

```
int i;
for( i=10; i>=1; i--){
    printf("I am good");
}
```

Decrement

Iteration

10 >= 1

9 >= 1

8 >= 1

⋮

1 >= 1

0 >= 1 ← exit

Decremental for loop

--i is also allowed

Slide



# For loop

```
int i;
for (i=1; i<10; (i++)) {
    printf ("I am best");
}
```

Relational

Iteration 1 < 10

Iteration 2 < 10 }

⋮

Iteration 9 < 10

Iteration 10 < 10 ← exit condition

9 times

Slide



# For loop

No. of times for loop will execute

```
int i;
for(i=1; i<=100; i++){
    printf("%d", i);
}
```

No. of times loop execute

100

lb ← Initial value of i

ub. Last value Condition true

lb = 1

ub = 100

Slide



# For loop

```
int i
for( i=1; i<100; i++){
    printf("%d", i);
}
```

99 times

value printed.

i

Lb = 1

ub = 99

=

```
int i
for (i=10; i<=100; i++){
    printf("I am good");
}
```

ub - Lb + 1

100 - 10 + 1

91 times

0 — 5

0, 1, 2, 3, 4, 5

1 - 5

1, 2, 3, 4, 5

Slide



# For loop

Ub - Lb + 1 ,

int i  i++  Incremental

[A] for (i = 23; i <= 63; i++)
    printf ("I am good");

63 - 23 + 1 = 41 times

[B] for (i = 60; i <= 100; i++)
    printf ("I am good");

100 - 60 + 1
40 + 1 = 41

if increment by 1

[C] for (i = 10; i <= 1; i++)
    printf ("I am good");

10 <= 1

Ans is 0

[D] for (i = 10; i < 79; i++)
    printf ("I am good");

78 - 10 + 1
69 times

Slide



# For loop

UB

LB

```
int i
for (i=111; i>=3; i--)
    printf("I am good");
```

* i--

only i-- and i++

UB - LB + 1

111 - 3 + 1

109

Slide



# For loop

Is expr1 , expr2 , ~~expr~~ expr3 is mandatory

Can we skip them

```
int c=10;
for ( ; i<=15; i++)
    printf ("I am good");
```

yes it will execute properly

6 times

expression is Not present

but we have to put semicolon.

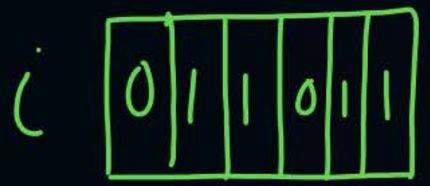
Slide



# For loop

```
int i;
for( ; i<=15; i++){
    printf("I am good");
}
```

depends upon garbage value Loop will behave.




Random value assigned in previous use of memory will be present.

Garbage (कचरा)

Slide



# For loop

```
int i = 1;
for ( ; i <= 10; ) {
    printf("I am good");
}
```

Initialization, Condition

1 time, every time

Iteration 1 — $1 <= 10$ — print

Iteration 2 — $1 <= 10$ — print

Iteration 3 — $1 <= 10$ — print

.

.

.

Infinite time

Slide

```
int i=1;
for(; i<=3 ; printf("yes"))
    printf("No");
    i=i+1;
}
```

i=2

i=3

i=4

Iteration 1; 1<=3. No yes

Iteration 2: 2<=3 No yes

Iteration 3<=3 No yes

Iteration 4 4<=3

functions can be allowed in

{ expr1, expr2

expr3



# For loop

```
int i = 1;
for( ;  ; i++){
    printf("I am good");
}
```

Answer

Infinite times printing

exp2 को eliminate

if exp2 is empty then it is always evaluates to true

Slide



# For loop

Infinile loop runs the program without stop

Slide



# Question

#Q. Consider the following program

```
#include<stdio.h>
int main(){
        int i= -1;
        for (; ++i; i++){
                printf("I am a good Student");
        }
        return 0 ;
}
```

Number of times loop is E

A. 0
B. 1
C. infinite
D. 2

expr

-1

++i

Increment, karega Q

++i

0 ← false

loop



# Question

8 bits  0 - 255

$256 = 2^8$

#Q. Consider the following program

```
#include<stdio.h>
    int main (){
    char ch ;
     for(ch=1;ch;ch++)
         printf("%c\n",ch);
     return 0;
}
```

Ch = 1

1, 2, 3

255

What is TRUE about this program when we run the program

Number of times loop is Executed

A. Compiler Error

B. Warning

C. Loop will run for 1 time

D. Loop will run for 255 times




1 2 127

Slide



# GATE 2015

#Q. consider the folllowing program

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k=k-(--j);
    for (i=0; i<5; i++)          {
        switch(i+k)              {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

The Number of times printf executed is____

Slide



# 2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

for loop

exp$_1$, exp$_2$, exp$_3$ timing

No. of times loop execule.

Slide

# THANK - YOU

